Character Creation Program:

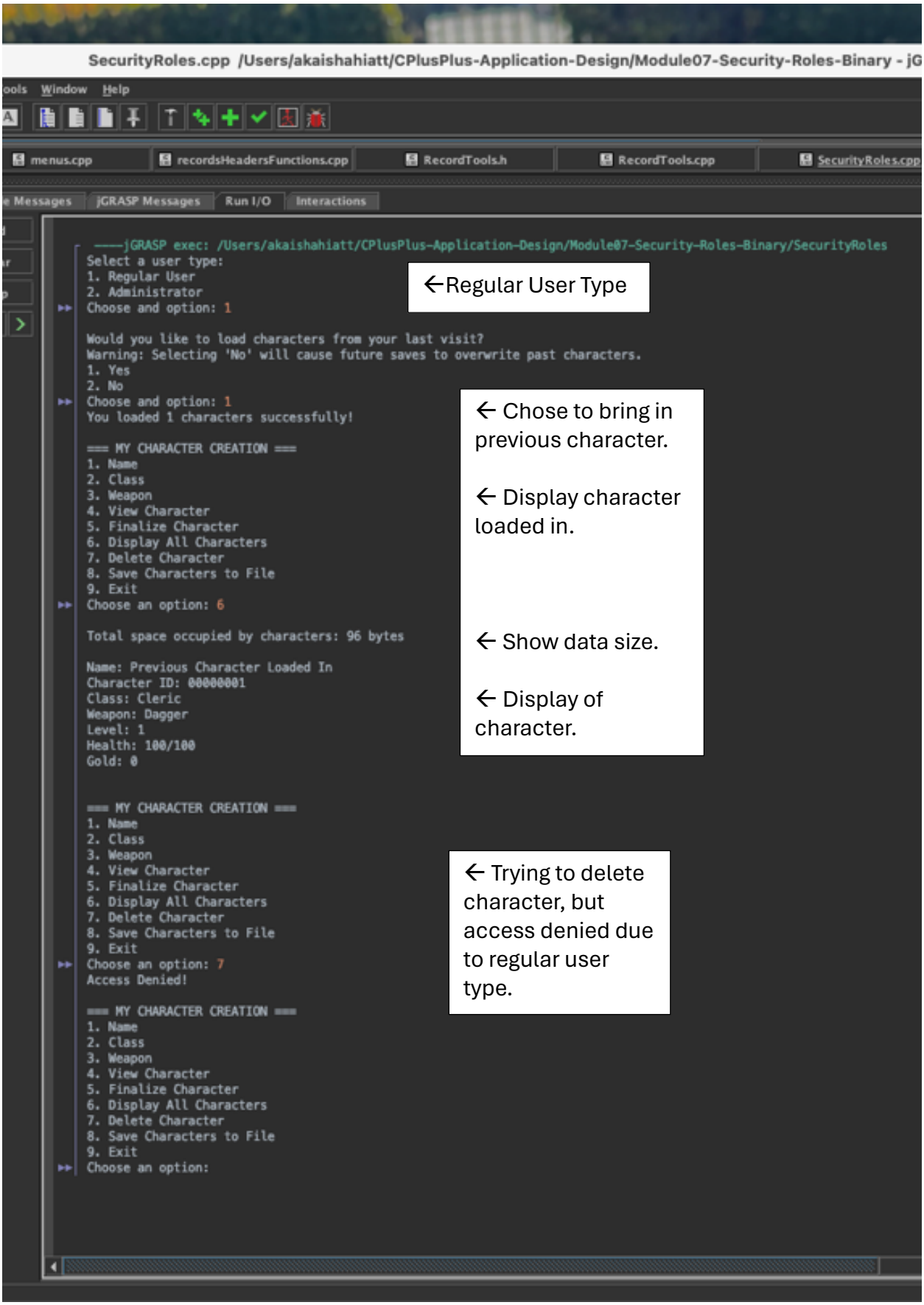
This program allows the user to create fantasy characters with multiple attributes and save them to a file for future use. It starts by asking the user if they are a regular user, or an administrator. The only difference between the two is that the administrator has the authority to delete characters. It then asks the user if they would like to load characters from their last visit. If they select yes, the characters from the last save are loaded, if no, then they aren't, and future saves will overwrite the previous data. The user is then taken to a main menu. The first three options allow them to select the attributes of a character currently in creation. If they do not select anything, those attributes will simply say [unknown]. Then there is an option to view the character you are currently creating. This prints their selections to the screen. They are then able to finalize the character, which officially adds the character to an array using a constructor which takes their selections, as well as adds some preliminary default attributes, such as 100/100 health, and 0 gold. It also assigns a character ID, which will never be reused, even if that character is deleted, unless the program is run without loading a previous save. Once a character has been finalized, the first three options in the main menu will now create a new character. The user can create up to 5 characters. There is also an option to display characters, which will print a list of your characters to the screen, including the default attributes assigned in the constructor, and the ID. There is an option available to administrators which allows someone to delete a character from the array, given that they know the ID number. The function uses binary sort to find the character, since they are stored in order by ID, and delete that character. Finally, there is an option to save to file. This option will overwrite the previous save and store the characters to a .txt file.

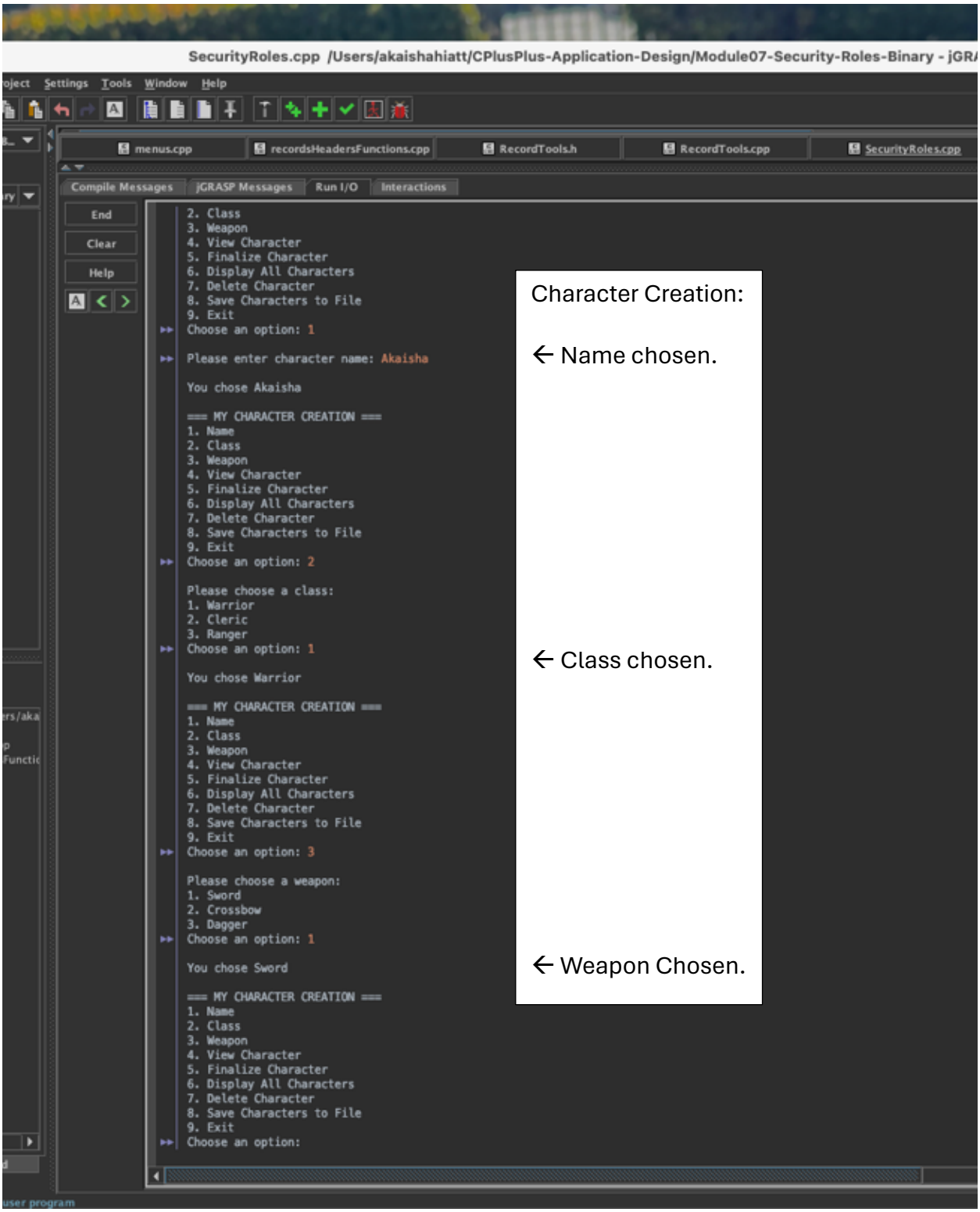
Concepts Demonstrated:

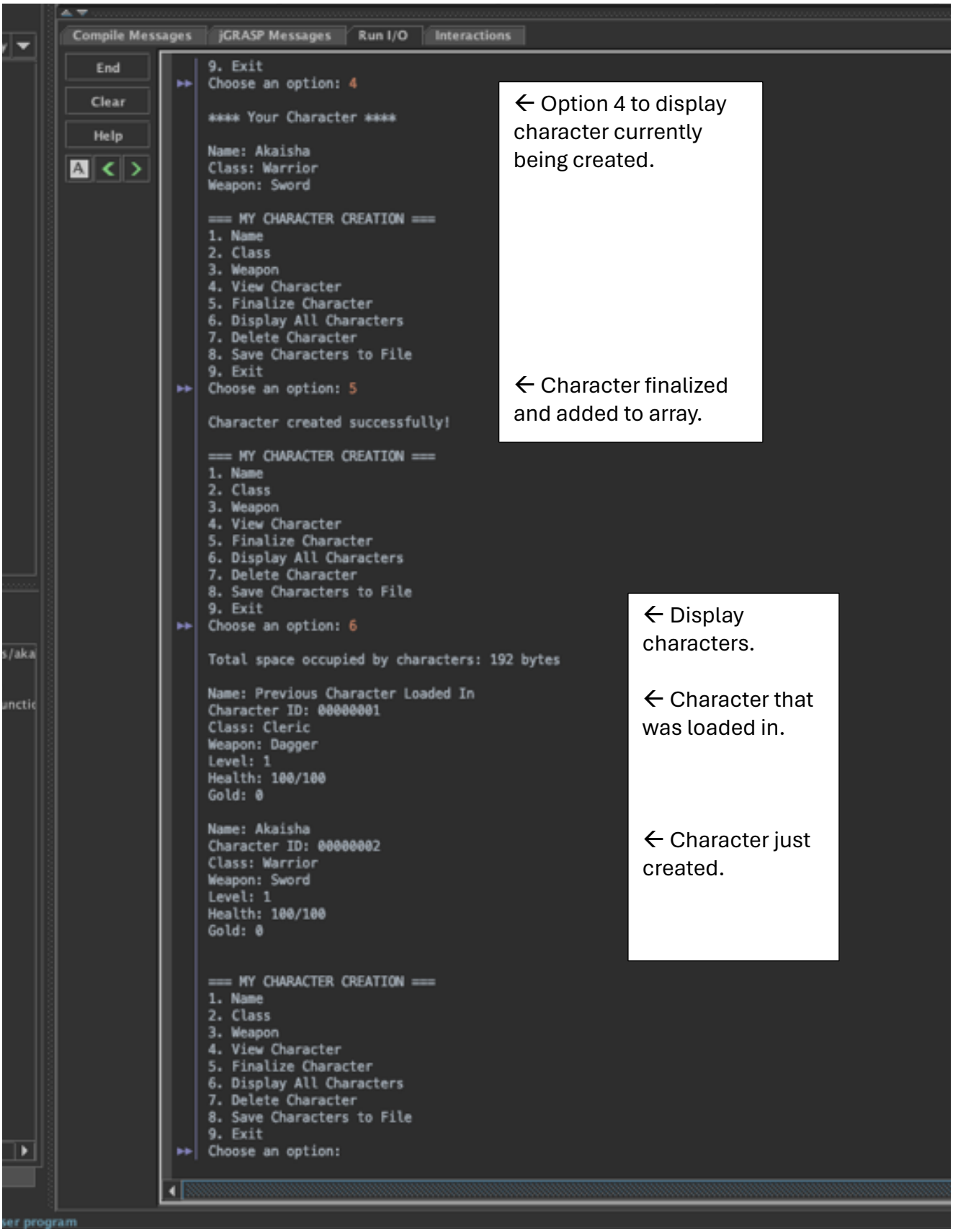
1. **Data types and data sizes:** This program uses several different data types, such as int (id number, health, level, etc.), double(gold), string(character name, class, weapon, etc.), bool(used in admin and user classes to determine whether the user has access to each function). The program also works with data sizes by using sizeof() to determine the size of a character, and display how many bytes are being occupied by the created characters.
2. **File Types:** This program uses a .txt file to save character information so that it can be reloaded to the array the next time the program is run. The character information is separated by commas, allowing it to be easily reassigned to the correct data types and used to recreate the characters using a special constructor.
3. **Use of pointers:** A pointer called currentUser is used to determine whether or not the user is a regular user, or an admin, and then determine whether they are able to carry out certain functions, specifically, deletion of characters.
4. **Arrays:** This program uses an array to store up to 5 characters, and subsequently display them or save them to a file. It utilizes a int variable characterCount to keep track of the amount of characters stored, and shifts the characters to the

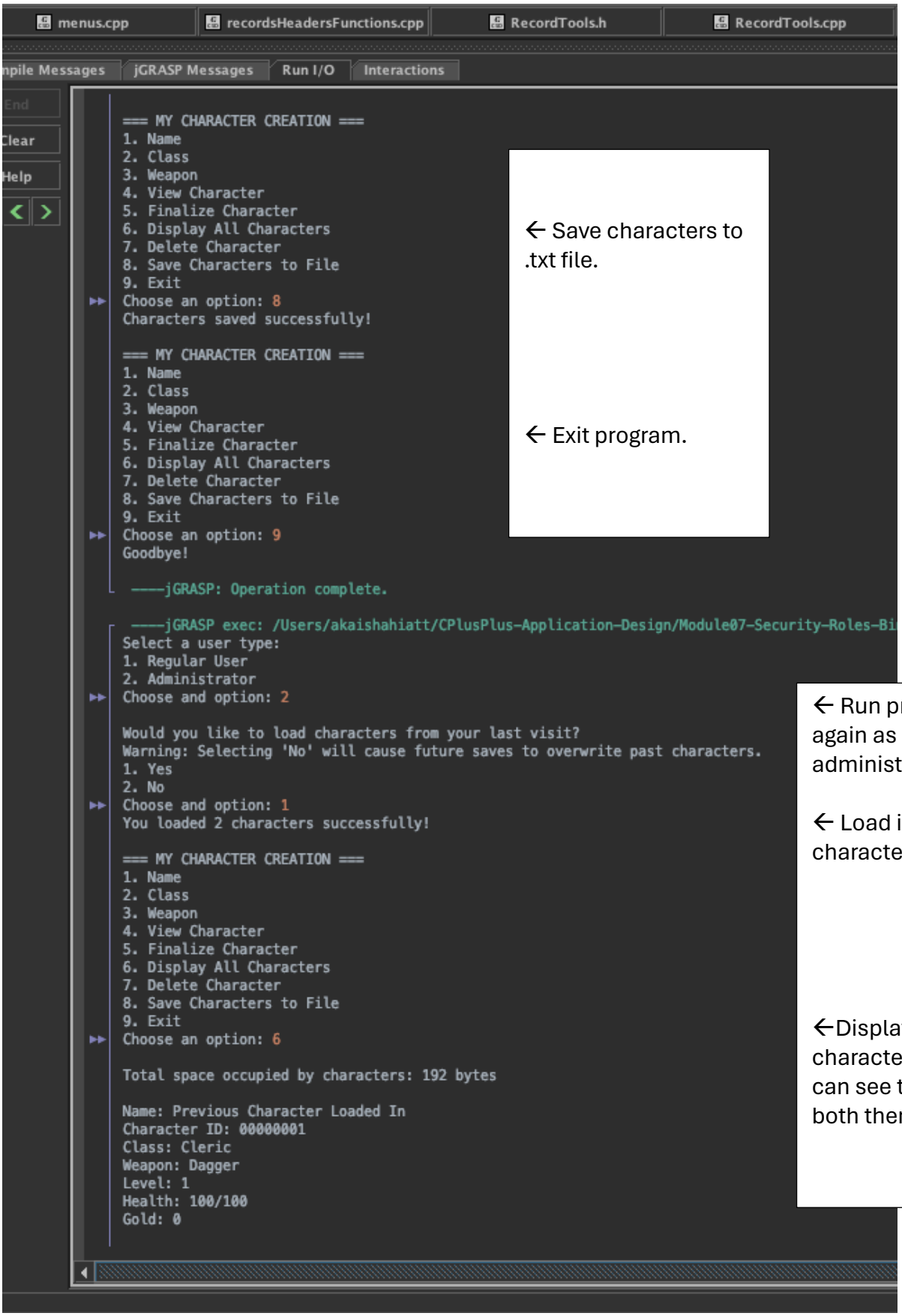
left when one is deleted, allowing for more to be added, and maintaining the correct order based on ID number.

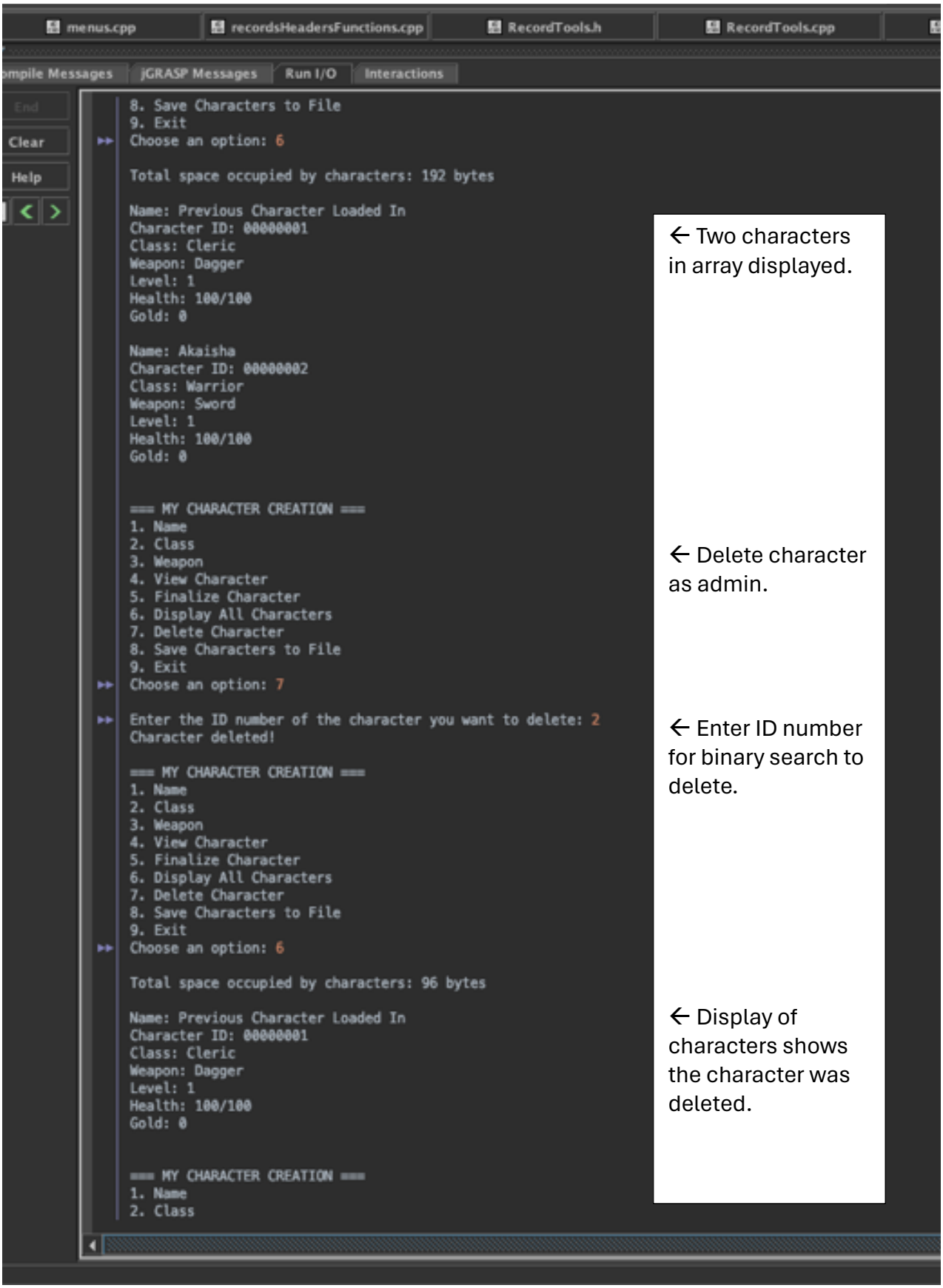
5. **Binary Search:** This program uses a binary search to delete characters by asking the user to enter the ID number of the character they would like to delete. Since they are in order by ID number, a binary search is an effective way to find the character so that it can be deleted.
6. **Use of Strings:** There are many strings used throughout the program to store user input and character attributes.
7. **File I/O:** This program uses fstream library to read information from and write information to a .txt file. It also converts that information back to its correct variable types, and then uses a special constructor to recreate the original array.
8. **At least two classes:** There are three classes used in this program. Admin, User, and Character. The Admin and User classes are used to create a security level for the program, while the character class is used to create character objects and store them in an array of characters.

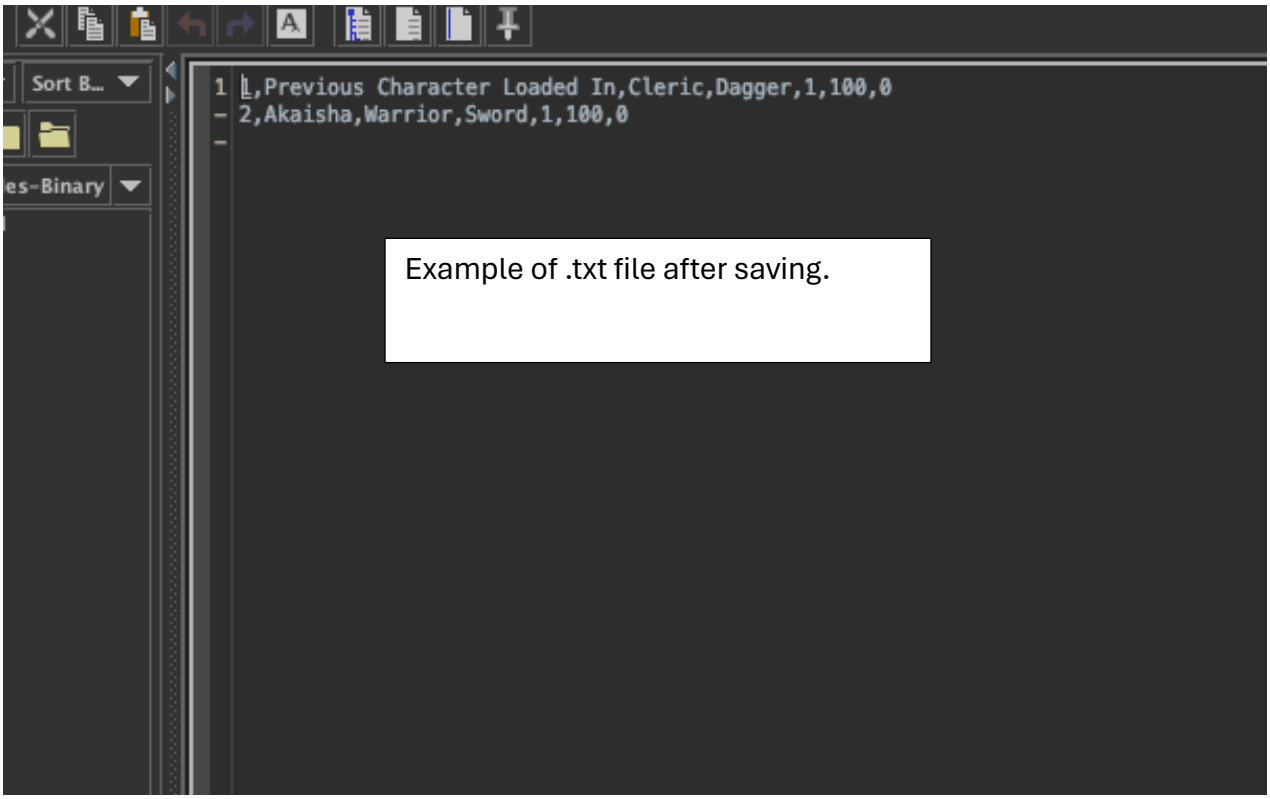

SecurityRoles.cpp /Users/akaishahiatt/CPlusPlus-Application-Design/Module07-Security-Roles-Binary - jG
Window Help
menus.cpp
recordsHeadersFunctions.cpp
RecordTools.h
RecordTools.cpp
SecurityRoles.cpp
jGRASP Messages
Run I/O
Interactions
----jGRASP exec: /Users/akaishahiatt/CPlusPlus-Application-Design/Module07-Security-Roles-Binary/SecurityRoles
Select a user type:
1. Regular User
2. Administrator
Choose and option: 1
Would you like to load characters from your last visit?
Warning: Selecting 'No' will cause future saves to overwrite past characters.
1. Yes
2. No
Choose and option: 1
You loaded 1 characters successfully!
=== MY CHARACTER CREATION ===
1. Name
2. Class
3. Weapon
4. View Character
5. Finalize Character
6. Display All Characters
7. Delete Character
8. Save Characters to File
9. Exit
Choose an option: 6
Total space occupied by characters: 96 bytes
Name: Previous Character Loaded In
Character ID: 00000001
Class: Cleric
Weapon: Dagger
Level: 1
Health: 100/100
Gold: 0
=== MY CHARACTER CREATION ===
1. Name
2. Class
3. Weapon
4. View Character
5. Finalize Character
6. Display All Characters
7. Delete Character
8. Save Characters to File
9. Exit
Choose an option: 7
Access Denied!
=== MY CHARACTER CREATION ===
1. Name
2. Class
3. Weapon
4. View Character
5. Finalize Character
6. Display All Characters
7. Delete Character
8. Save Characters to File
9. Exit
Choose an option:
←Regular User Type
← Chose to bring in previous character.
← Display character loaded in.
← Show data size.
← Display of character.
← Trying to delete character, but access denied due to regular user type.


SecurityRoles.cpp /Users/akaishahiatt/CPlusPlus-Application-Design/Module07-Security-Roles-Binary - jGRA
Settings Tools Window Help
menus.cpp recordsHeadersFunctions.cpp RecordTools.h RecordTools.cpp SecurityRoles.cpp
Compile Messages jGRASP Messages Run I/O Interactions
End
Clear
Help
2. Class
3. Weapon
4. View Character
5. Finalize Character
6. Display All Characters
7. Delete Character
8. Save Characters to File
9. Exit
Choose an option: 1
Please enter character name: Akaisha
You chose Akaisha
=== MY CHARACTER CREATION ===
1. Name
2. Class
3. Weapon
4. View Character
5. Finalize Character
6. Display All Characters
7. Delete Character
8. Save Characters to File
9. Exit
Choose an option: 2
Please choose a class:
1. Warrior
2. Cleric
3. Ranger
Choose an option: 1
You chose Warrior
=== MY CHARACTER CREATION ===
1. Name
2. Class
3. Weapon
4. View Character
5. Finalize Character
6. Display All Characters
7. Delete Character
8. Save Characters to File
9. Exit
Choose an option: 3
Please choose a weapon:
1. Sword
2. Crossbow
3. Dagger
Choose an option: 1
You chose Sword
=== MY CHARACTER CREATION ===
1. Name
2. Class
3. Weapon
4. View Character
5. Finalize Character
6. Display All Characters
7. Delete Character
8. Save Characters to File
9. Exit
Choose an option:
Character Creation:
← Name chosen.
← Class chosen.
← Weapon Chosen.


Compile Messages
jGRASP Messages
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End
Clear
Help
9. Exit
Choose an option: 4
**** Your Character ****
Name: Akaisha
Class: Warrior
Weapon: Sword
=== MY CHARACTER CREATION ===
1. Name
2. Class
3. Weapon
4. View Character
5. Finalize Character
6. Display All Characters
7. Delete Character
8. Save Characters to File
9. Exit
Choose an option: 5
Character created successfully!
=== MY CHARACTER CREATION ===
1. Name
2. Class
3. Weapon
4. View Character
5. Finalize Character
6. Display All Characters
7. Delete Character
8. Save Characters to File
9. Exit
Choose an option: 6
Total space occupied by characters: 192 bytes
Name: Previous Character Loaded In
Character ID: 00000001
Class: Cleric
Weapon: Dagger
Level: 1
Health: 100/100
Gold: 0
Name: Akaisha
Character ID: 00000002
Class: Warrior
Weapon: Sword
Level: 1
Health: 100/100
Gold: 0
=== MY CHARACTER CREATION ===
1. Name
2. Class
3. Weapon
4. View Character
5. Finalize Character
6. Display All Characters
7. Delete Character
8. Save Characters to File
9. Exit
Choose an option:
← Option 4 to display character currently being created.
← Character finalized and added to array.
← Display characters.
← Character that was loaded in.
← Character just created.


menus.cpp
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=== MY CHARACTER CREATION ===
1. Name
2. Class
3. Weapon
4. View Character
5. Finalize Character
6. Display All Characters
7. Delete Character
8. Save Characters to File
9. Exit
Choose an option: 8
Characters saved successfully!
=== MY CHARACTER CREATION ===
1. Name
2. Class
3. Weapon
4. View Character
5. Finalize Character
6. Display All Characters
7. Delete Character
8. Save Characters to File
9. Exit
Choose an option: 9
Goodbye!
----jGRASP: Operation complete.
----jGRASP exec: /Users/akaishahiatt/CPlusPlus-Application-Design/Module07-Security-Roles-Bi
Select a user type:
1. Regular User
2. Administrator
Choose and option: 2
Would you like to load characters from your last visit?
Warning: Selecting 'No' will cause future saves to overwrite past characters.
1. Yes
2. No
Choose and option: 1
You loaded 2 characters successfully!
=== MY CHARACTER CREATION ===
1. Name
2. Class
3. Weapon
4. View Character
5. Finalize Character
6. Display All Characters
7. Delete Character
8. Save Characters to File
9. Exit
Choose an option: 6
Total space occupied by characters: 192 bytes
Name: Previous Character Loaded In
Character ID: 00000001
Class: Cleric
Weapon: Dagger
Level: 1
Health: 100/100
Gold: 0
← Save characters to .txt file.
← Exit program.
← Run program again as administrator.
← Load in characters.
←Display characters so we can see them both there.


menus.cpp
recordsHeadersFunctions.cpp
RecordTools.h
RecordTools.cpp
Compile Messages
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Help
8. Save Characters to File
9. Exit
Choose an option: 6
Total space occupied by characters: 192 bytes
Name: Previous Character Loaded In
Character ID: 00000001
Class: Cleric
Weapon: Dagger
Level: 1
Health: 100/100
Gold: 0
Name: Akaisha
Character ID: 00000002
Class: Warrior
Weapon: Sword
Level: 1
Health: 100/100
Gold: 0
=== MY CHARACTER CREATION ===
1. Name
2. Class
3. Weapon
4. View Character
5. Finalize Character
6. Display All Characters
7. Delete Character
8. Save Characters to File
9. Exit
Choose an option: 7
Enter the ID number of the character you want to delete: 2
Character deleted!
=== MY CHARACTER CREATION ===
1. Name
2. Class
3. Weapon
4. View Character
5. Finalize Character
6. Display All Characters
7. Delete Character
8. Save Characters to File
9. Exit
Choose an option: 6
Total space occupied by characters: 96 bytes
Name: Previous Character Loaded In
Character ID: 00000001
Class: Cleric
Weapon: Dagger
Level: 1
Health: 100/100
Gold: 0
=== MY CHARACTER CREATION ===
1. Name
2. Class
← Two characters in array displayed.
← Delete character as admin.
← Enter ID number for binary search to delete.
← Display of characters shows the character was deleted.


Sort B...
es-Binary
1 1,Previous Character Loaded In,Cleric,Dagger,1,100,0
2,Akaisha,Warrior,Sword,1,100,0
Example of .txt file after saving.